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# =====
# 3-STOREY MASONRY FRAME – DEBUG VERSION
# Same as the original script, with:
# - catch blocks around every analysis phase
# - verbose output at each step
# - fixes applied for the three critical issues found in review
#   (settlement DOF, floor load formula, pushover fallback)
# =====

proc phase {title} {
    puts "\n=====
    puts " $title"
    puts "=====
}

proc check {label result} {
    if {$result == 0} {
        puts " \[OK\] $label"
    } else {
        puts " \[FAIL\] $label (return code: $result)"
    }
}

# =====
# SETUP
# =====

phase "MODEL SETUP"

if {[catch {wipe} err]} { puts " wipe: $err" }
if {[catch {model basic -ndm 3 -ndf 6} err]} {
    puts " FATAL – model setup failed: $err"
    return
}
puts " model basic -ndm 3 -ndf 6 : OK"

set H_story 3.0
set L_span 3.0
set L_pier 1.0
set T_pier 0.25
set H_span 0.80

set E 1700.0e6
set G 550.0e6
set fc 6.0e6
set c 0.150e6
set Gc 6.0
set mu0 0.40
set beta 0.30

set rho 1200.0
set g 9.81

# =====
# NODES
# =====

phase "NODES"

if {[catch {
    node 1 0.0 0.0 0.0
    node 2 $L_span 0.0 0.0

    node 3 0.0 0.0 $H_story
    node 4 $L_span 0.0 $H_story

    node 5 0.0 0.0 [expr 2.0*$H_story]
    node 6 $L_span 0.0 [expr 2.0*$H_story]

    node 7 0.0 0.0 [expr 3.0*$H_story]
    node 8 $L_span 0.0 [expr 3.0*$H_story]

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node 9    0.0      0.0    [expr 0.5*$H_story]
node 10   $L_span  0.0    [expr 0.5*$H_story]
node 11   0.0      0.0    [expr 1.5*$H_story]
node 12   $L_span  0.0    [expr 1.5*$H_story]
node 13   0.0      0.0    [expr 2.5*$H_story]
node 14   $L_span  0.0    [expr 2.5*$H_story]

node 15   [expr $L_span/2.0] 0.0 $H_story
node 16   [expr $L_span/2.0] 0.0 [expr 2.0*$H_story]
node 17   [expr $L_span/2.0] 0.0 [expr 3.0*$H_story]

puts " All 17 nodes created : OK"

} err]] {
    puts " FATAL - node creation failed: $err"
    return
}

# =====
# BOUNDARY CONDITIONS
# FIX #1 (from review): node 1 DOF 3 (vertical) left FREE for settlement
# =====

phase "BOUNDARY CONDITIONS"

if {[catch {
    fix 1 1 1 0 1 1 1    ;# vertical DOF (3) free - needed for settlement
    fix 2 1 1 1 1 1 1
    puts " node 1 : fixed in X Y Rx Ry Rz - vertical (Z) FREE for settlement"
    puts " node 2 : fully fixed"
} err]] {
    puts " FATAL - boundary conditions failed: $err"
    return
}

# =====
# ELEMENTS - PIERS
# =====

phase "PIER ELEMENTS"

foreach {eid ni nj nk} {
    1 1 3 9
    2 2 4 10
    3 3 5 11
    4 4 6 12
    5 5 7 13
    6 6 8 14
} {
    if {[catch {
        element Macroelement3d $eid \
            $ni $nj $nk \
            0.0 0.0 1.0 \
            0.0 1.0 0.0 \
            -tremuri \
            $H_story $L_pier $T_pier \
            $E $G $fc $mu0 $c $Gc $beta \
            -density $rho \
            -cmass \
            -pDelta
        puts " Pier element $eid (nodes $ni-$nj-$nk) : OK"
    } err]] {
        puts " FAILED - pier element $eid: $err"
    }
}

# =====
# ELEMENTS - SPANDRELS
# =====

phase "SPANDREL ELEMENTS"

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foreach {eid ni nj nk} {
    7  3  4  15
    8  5  6  16
    9  7  8  17
} {
    if {[catch {
        element Macroelement3d $eid \
            $ni $nj $nk \
            1.0 0.0 0.0 \
            0.0 1.0 0.0 \
            -tremuri \
            $H_span $L_span $T_pier \
            $E $G $fc $mu0 $c $Gc $beta \
            -density $rho \
            -cmass \
            -pDelta
        puts " Spandrel element $eid (nodes $ni-$nj-$nk) : OK"
    } err]} {
        puts " FAILED - spandrel element $eid: $err"
    }
}

# =====
# RECORDERS
# =====

phase "RECORDERS"

recorder Node -file RoofDisp.out      -time -node 8      -dof 1 disp
recorder Node -file SupportReaction.out -time -node 1 2 -dof 1 reaction
recorder Element -file PierForces.out  -time -ele 1 2 3 4 5 6 basicForce
recorder Element -file SpanForces.out  -time -ele 7 8 9      basicForce
puts " Recorders defined : OK"

# =====
# GRAVITY LOADS
# FIX #2 (from review): floor load formula corrected
# Original: -5.0*$g*$rho*$L_span*$T_pier  ( $\approx 44\,100\text{ N}$  - ~12x too large)
# Corrected: -5.0e3*$L_span*$T_pier      (=  $3\,750\text{ N}$ , i.e.  $5\text{ kN/m}^2$  floor load)
# If the intent was masonry self-weight use: - $\rho * g * H_{\text{story}} * L_{\text{span}} * T_{\text{pier}}$ 
# =====

phase "GRAVITY LOADS"

set floorLoad [expr -5.0e3 * $L_span * $T_pier]
puts " Floor load per node : $floorLoad N  ( $5\text{ kN/m}^2 \times L_{\text{span}} \times T_{\text{pier}}$ )"
puts " Original formula would have given: [expr -5.0*$g*$rho*$L_span*$T_pier] N"

pattern Plain 10 Linear {
    for {set e 1} {$e <= 9} {incr e} {
        eleLoad -ele $e -type -selfWeight 0.0 0.0 [expr -$g]
    }
    load 3 0.0 0.0 $floorLoad 0.0 0.0 0.0
    load 4 0.0 0.0 $floorLoad 0.0 0.0 0.0
    load 5 0.0 0.0 $floorLoad 0.0 0.0 0.0
    load 6 0.0 0.0 $floorLoad 0.0 0.0 0.0
    load 7 0.0 0.0 $floorLoad 0.0 0.0 0.0
    load 8 0.0 0.0 $floorLoad 0.0 0.0 0.0
}

system BandGeneral
numberer Plain
constraints Transformation
test NormUnbalance 100.0 50 5
algorithm Newton
integrator LoadControl 0.01
analysis Static

puts " Applying gravity loads (100 steps)..."
set ok [analyze 100]
check "Gravity analysis" $ok

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if {$ok != 0} {
    puts " FATAL – gravity analysis failed. Check material parameters and element setup."
    return
}

loadConst -time 0.0
puts " Gravity loads held constant. Pseudo-time reset to 0."

# =====
# FOUNDATION SETTLEMENT
# node 1, DOF 3 (Z) is now free – sp command will work correctly
# =====

phase "FOUNDATION SETTLEMENT"

pattern Plain 20 Linear {
    sp 1 3 -0.001
}

integrator LoadControl 0.01

puts " Applying settlement (200 steps → 2 mm at node 1)..."
set ok [analyze 200]
check "Settlement analysis" $ok

if {$ok != 0} {
    puts " WARNING – settlement analysis did not converge cleanly."
    puts " Continuing with current state, but results may be affected."
}

loadConst -time 0.0
record
puts " Settlement held constant. Pseudo-time reset to 0."

# =====
# PUSHOVER LOAD PATTERN
# =====

phase "PUSHOVER SETUP"

pattern Plain 30 Linear {
    load 3 0.20 0.0 0.0 0.0 0.0 0.0
    load 4 0.20 0.0 0.0 0.0 0.0 0.0
    load 5 0.40 0.0 0.0 0.0 0.0 0.0
    load 6 0.40 0.0 0.0 0.0 0.0 0.0
    load 7 0.60 0.0 0.0 0.0 0.0 0.0
    load 8 0.60 0.0 0.0 0.0 0.0 0.0
}
puts " Triangular load pattern defined (ratios 1:2:3 per floor)"

constraints Transformation
numberer Plain
system BandGeneral
test NormUnbalance 100.0 50 5
algorithm Newton
analysis Static

set targetDisp 0.20
set incr 0.0001
set nSteps [expr int($targetDisp/$incr)]

integrator DisplacementControl 8 1 $incr

puts " Target displacement : $targetDisp m"
puts " Increment : $incr m"
puts " Number of steps : $nSteps"

# =====
# PUSHOVER ANALYSIS
# FIX #3 (from review): step-by-step loop – no overshoot on fallback
# =====

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phase "PUSHOVER ANALYSIS"

set completedSteps 0
set usingInitial 0

while {$completedSteps < $nSteps} {

    set ok [analyze 1]

    if {$ok == 0} {
        incr completedSteps
        # Revert to standard Newton once we're past a hard spot
        if {$usingInitial} {
            algorithm Newton
            test NormUnbalance 100.0 50 5
            set usingInitial 0
        }
    } else {
        if {!$usingInitial} {
            puts " Newton failed at step $completedSteps – switching to Newton -initial"
            test NormUnbalance 100.0 200 5
            algorithm Newton -initial
            set usingInitial 1
        }
        set ok [analyze 1]
        if {$ok == 0} {
            incr completedSteps
        } else {
            puts " STOP – both algorithms failed at step $completedSteps / $nSteps"
            puts " Roof displacement reached: [expr $completedSteps * $incr] m"
            break
        }
    }
}

# Progress report every 500 steps
if {[expr $completedSteps % 500] == 0 && $completedSteps > 0} {
    puts " ... step $completedSteps / $nSteps \
    (disp ≈ [expr $completedSteps * $incr] m)"
}

puts ""
puts " Pushover completed: $completedSteps of $nSteps steps"
puts " Final roof displacement: [expr $completedSteps * $incr] m"
puts " Results written to: RoofDisp.out, SupportReaction.out"
puts " Element results : PierForces.out, SpanForces.out"

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